Project ClearRiver: Real-Time Kinetic Analysis of Bio-char Adsorption in Resource-Constrained Industrial Settings

An Integrated IoT Spectrophotometric and Automated Compliance Framework for Small-Scale Textile Wastewater Treatment in Bangladesh

Tasfiea Rubaiea
Lamyea Zaman Sneha
Anika Tabassum Adiba
and Khadija Tul Kubra
Independent Research, Dinajpur Govt. Women's College, Dinajpur, Bangladesh

Abstract. Unregulated discharge of synthetic textile dyes into aquatic ecosystems poses a severe environmental crisis in developing industrial nations, frequently producing hypoxic conditions in vital waterways. Simultaneously, the agricultural sector generates millions of tonnes of lignocellulosic waste—primarily rice husks—that are routinely incinerated, compounding air-quality degradation. This study presents the design, construction, and experimental validation of an ultra-low-cost ($\leq$1000 BDT) continuous-flow bio-filtration system integrated with a custom Internet-of-Things (IoT) spectrophotometric array. Locally sourced *Oryza sativa* husks from Dinajpur, Bangladesh, were carbonised at 400 °C under anaerobic conditions to yield a highly porous bio-char adsorbent that captures organic dye molecules via physisorption and π–π dispersion forces. Real-time monitoring was achieved using a Time-Division-Multiplexed (TDM) RGB optical array driven by an ATmega328P microcontroller, calibrated against a five-point Beer–Lambert baseline ($R^2 = 0.992$). Triplicate continuous-flow trials ($n = 3$) demonstrated a mean pollutant-mass reduction of **93%** over 300 s, with effluent clarity surpassing the 85 % regulatory threshold at $t \approx 252$ s. A first-order adsorption rate constant of $k = (0.0089 \pm 0.0003)$/s was extracted in real time. Equilibrium behaviour was characterised using Langmuir and Freundlich isotherm models, with the latter ($R^2 = 0.978$) providing a superior fit consistent with the heterogeneous pore-size distribution of rice-husk bio-char. Sensor data are transmitted via a Node.js serial bridge to a Next.js dashboard that performs asynchronous kinetic calculus and autonomously generates tamper-evident PDF compliance reports. This work establishes a scalable, decentralised paradigm for industrial-grade wastewater analytics accessible to small and medium enterprises (SMEs) at minimal capital expenditure.

Keywords: bio-char adsorption; textile wastewater; Beer–Lambert law; IoT spectrophotometry; first-order kinetics; Langmuir isotherm; Freundlich isotherm; environmental compliance; rice husk; Bangladesh; SME.

1. Introduction

1.1 Environmental Context and Motivation

The garment manufacturing sector contributes over 80% of Bangladesh's national export revenue, yet its environmental footprint is severe. Untreated effluent containing toxic, non-biodegradable synthetic dyes is routinely discharged into major waterways; rivers such as the Buriganga have experienced catastrophic drops in dissolved oxygen as high dye concentrations attenuate the sunlight required for aquatic photosynthesis [1]. This degradation disproportionately affects small and medium enterprises

(SMEs), which collectively dominate production yet lack access to commercially viable continuous wastewater treatment.

Concurrently, northern Bangladesh generates massive quantities of rice husks as an agricultural processing byproduct. These are overwhelmingly disposed of through open-air incineration, exacerbating particulate air-quality problems while neglecting their documented potential as high-value precursors for chemical adsorbents [2].

1.2 Research Gaps

Despite an established body of literature on bio-char adsorption and low-cost optical sensing, four barriers to practical SME deployment persist:

i. **Economic barriers.** Commercial spectrophotometers cost thousands of dollars, creating a financial incentive for illegal effluent discharge.

ii. **Absence of continuous monitoring.** Academic bio-char implementations rarely integrate real-time performance verification; a filter without instrumentation provides no regulatory evidence.

iii. **Fragmented kinetic analysis.** Although first-order kinetic decay theory is well-established, no accessible software architecture exists to solve the governing differential equations asynchronously from live sensor data.

iv. **Lack of automated compliance reporting.** No standardised, ultra-low-cost framework bridges raw sensor readings and verifiable, autonomously generated environmental compliance documents.

1.3 Research Question and Hypotheses

To what extent can an integrated, decentralised system—utilising rice-husk-derived bio-char and a TDM IoT optical array—provide a precise and scalable alternative for real-time kinetic modelling and autonomous environmental compliance monitoring in resource-constrained industrial settings?

Three testable hypotheses are advanced:

H_1 **(Adsorption Efficiency).** Anaerobic pyrolysis of *Oryza sativa* husks at ≈400 °C produces bio-char sufficient to achieve >90 % pollutant-mass reduction in continuous-flow configuration.

H_2 **(Sensor Fidelity).** An RGB-multiplexed LDR array governed by the Beer–Lambert Law can isolate $\lambda_{\max}$ dynamically and serve as a high-fidelity, ultra-low-cost proxy for commercial spectrophotometry.

H_3 **(Automation).** A cloud-integrated state manager can integrate first-order kinetic differential equations in real time and autonomously generate verifiable compliance reports once effluent clarity exceeds 85 % continuously.

1.4 Novelty

Unlike prior literature treating bio-char filtration and DIY sensing as independent disciplines, this work delivers an end-to-end "Wastewater-to-Compliance" pipeline—from biomass carbonisation through real-time kinetic modelling to autonomous cryptographic compliance reporting—for under 1000 BDT, democratising industrial-grade monitoring for resource-constrained SMEs.

2. Theoretical Background

2.1 Lignocellulosic Carbonisation Chemistry

Rice husks consist primarily of cellulose (∼35%), hemicellulose (∼25%), and lignin (∼20%), with silica and ash comprising the remainder [2]. Anaerobic thermal pyrolysis drives off volatile hemicellulose compounds and carbonises the lignin skeleton, producing a bio-char with: (i) dramatically increased specific surface area; (ii) a heterogeneous micro- to mesopore distribution; (iii) graphene-like aromatic sheets providing π–π interaction sites for aromatic dye molecules; and (iv) residual hydroxyl and carboxyl groups enabling hydrogen bonding.

2.2 Beer–Lambert Law

Dye concentration in the effluent stream is quantified via the Beer–Lambert Law:

$$A = \varepsilon \ell c \tag{1}$$

where A is the dimensionless absorbance, ε is the molar attenuation coefficient (L/(mol cm)), ℓ is the optical path length (cm), and c is the molar dye concentration (mol/L). The LDR array measures transmittance $T = I/I_0$; absorbance is recovered as:

$$A = -\log_{10}(T) \tag{2}$$

TDM cycling of red, green, and blue illumination enables dynamic identification of $\lambda_{\max}$ for the dye present, ensuring robust concentration tracking across varying effluent compositions [3, 4].

2.3 First-Order Kinetic Decay Model

Dye removal in the column follows the first-order differential equation [5]:

$$\frac{dC}{dt} = -kC \quad \Longrightarrow \quad C_t = C_0\, e^{-kt} \tag{3}$$

where k (1/s) is the adsorption rate constant. Linearising as $\ln(C_t/C_0) = -kt$ produces a diagnostic straight-line plot confirming first-order behaviour; the slope gives $-k$ directly.

2.4 Equilibrium Isotherm Models

The **Langmuir** model assumes monolayer adsorption on a homogeneous surface:

$$q_e = \frac{q_m K_L C_e}{1 + K_L C_e} \tag{4}$$

where q_e (mg/g) is equilibrium loading, q_m is the maximum monolayer capacity, K_L (L/mg) is the affinity constant, and C_e is the equilibrium solution concentration (mg/L).

The **Freundlich** model accounts for multilayer adsorption on heterogeneous surfaces:

$$q_e = K_F\, C_e^{1/n} \tag{5}$$

Values $1/n \in (0.1, 1.0)$ indicate favourable adsorption. Given the heterogeneous pore-size distribution inherent to lignocellulosic bio-char, the Freundlich model is expected to provide a superior fit [2].

3. Literature Review

3.1 Bio-char from Rice Husks

Rice-husk bio-char has been widely validated as a low-cost adsorbent for synthetic organic dyes [2]. Pyrolysis at 300 °C to 500 °C optimises the balance between specific surface area and retention of oxygenated functional groups. At 400 °C—adopted in this study—sufficient microporosity for rapid physisorption is achieved while retaining hydroxyl and carboxyl groups for targeted chemisorption of aromatic chromophores.

3.2 Low-Cost Spectrophotometric Sensing

Kaur and Singh [4] demonstrated that ATmega328P microcontrollers combined with LEDs and LDRs can replicate single-beam spectrophotometric measurement at a fraction of laboratory-grade cost, applying the Beer–Lambert relationship established by Ingle and Crouch [3]. This principle is directly applied here, extended with TDM multi-wavelength capability housed entirely within a sub-1000 BDT hardware budget.

3.3 First-Order Adsorption Kinetics

Ho and McKay [5] established the pseudo-first-order model as appropriate for systems where adsorbate concentration greatly exceeds available surface sites—a condition satisfied in a continuous-flow column where fresh polluted influent continuously replenishes the concentration gradient at the bio-char surface.

3.4 Research Gap and Positioning

Existing literature treats bio-char chemistry and IoT optical sensing as independent domains; no published work delivers a fully integrated pipeline from filtration through real-time kinetic modelling to autonomous compliance reporting. Project ClearRiver fills this gap with an end-to-end system within a total hardware budget below 1000 BDT.

4. Materials and Instrumentation

4.1 Adsorbent Preparation

4.1.1 Raw Material Sourcing

Raw *Oryza sativa* husks were obtained from agricultural milling facilities in Dinajpur, northern Bangladesh. Local sourcing reduces adsorbent cost to transport logistics alone.

4.1.2 Pre-Treatment

Husks were triple-rinsed with distilled water and oven-dried at 105 °C for 120 min to eliminate soluble impurities and residual moisture, ensuring that subsequent pyrolysis mass-loss measurements reflect volatile organic decomposition.

4.1.3 Anaerobic Pyrolysis

Dried husks were packed into a sealed metallic crucible with a single 2 mm vent to permit outgassing while maintaining anaerobic conditions. The crucible was externally heated for 30 min to a peak temperature of (400 ± 10) °C, monitored by a K-type thermocouple inserted through the vent. After heating, the bio-char was cooled to (25 ± 2) °C inside the sealed chamber to prevent auto-ignition and oxidative ash formation.

4.2 Continuous-Flow Column

The filter column comprised clear acrylic tubing (inner diameter 50 mm) packed in a vertical gradient, from bottom to top:

1. Cotton-wool exit particulate filter (10 mm depth).
2. 50 g activated rice-husk bio-char (80 mm depth).
3. Washed inert silica sand (30 mm depth) to distribute influent evenly across the bio-char bed.

A 10 mm inner-diameter clear acrylic tube routed from the column outlet served as the continuous-flow optical measurement cell. The column operated in gravity-fed, downward-flow mode regulated to 15 mL/min using a calibrated flow clamp.

4.3 IoT Optical Sensor Array

4.3.1 Housing

The sensor housing was fabricated from opaque black acrylic to eliminate ambient-light interference. The clear effluent tube bisects the housing, establishing an optical path length $\ell = 10$ mm.

4.3.2 Illumination

A common-cathode RGB LED (forward voltages: 2.0 V red, 3.2 V green and blue; drive current 20 mA) was mounted at 180° to the detector. Three 150 Ω current-limiting resistors were placed in series (one per channel) to protect the LED from the 5 V supply.

4.3.3 Photodetector

A GL5528 LDR (dark resistance ∼1 MΩ; illuminated resistance 10 kΩ to 100 kΩ) was wired in a voltage-divider with a 10 kΩ precision fixed resistor (±1 %):

$$V_{\text{out}} = V_{\text{in}} \cdot \frac{R_{\text{LDR}}}{R_{\text{LDR}} + R_{\text{fixed}}} \tag{6}$$

where $V_{\text{in}} = 5$ V. The output was fed to the microcontroller's analog input.

4.3.4 Microcontroller and TDM

An ATmega328P (16 MHz; 5 V) sequentially activated each LED channel with 100 ms inter-channel blanking to prevent optical cross-talk (TDM scheme). Each resulting analog voltage was digitised by the 10-bit ADC (resolution ≈4.88 mV/bit) and transmitted over USB serial at 9600 baud as a comma-separated RGB string.

4.4 Software Architecture

Three software tiers processed raw sensor data into compliance outputs.

Tier 1 (Middleware). A Node.js daemon polled the serial port, applied a five-point moving-average noise filter, converted ADC counts to transmittance via the calibration lookup table, and broadcast processed values via WebSockets.

Tier 2 (Dashboard). A Next.js/React application rendered multi-wavelength transmittance in real time and continuously refitted the first-order model (Equation 3) via weighted least-squares regression, updating the displayed k on each incoming data frame.

Tier 3 (Compliance Engine). Once $T_{\text{red}} \geq 85\,\%$ was sustained for a 10 s window, PDFKit compiled the kinetic plots, timestamped logs, final k, and a SHA-256 cryptographic hash of the data record into a tamper-evident PDF compliance report.

5. Experimental Methodology

5.1 Sensor Calibration

Prior to filtration trials, a five-point Beer–Lambert calibration was performed. Standard synthetic blue-dye solutions (0 mg/L, 10 mg/L, 25 mg/L, 40 mg/L and 50 mg/L) were prepared using Class-A volumetric glassware (±0.1 mL). The flow cell was first primed with distilled water to record the reference intensity I_0 ($T = 1.0$, $A = 0$). Each standard was pumped through the cell at the experimental flow rate; the steady-state ADC reading was recorded in triplicate. Absorbance was computed via Equation 2, and a linear regression $A = mc + b$ was fitted. An acceptance criterion of $R^2 > 0.98$ on the red channel ($\lambda \approx 620$ nm) was required before trials began.

5.2 Filtration Trials

Table 1 summarises the controlled experimental variables.

Table 1: Controlled experimental variables.

Variable	Value
Influent dye concentration	50 mg/L
Flow rate	15 mL/min
Operating temperature	$(25 \pm 2)\,°\text{C}$
Total volume processed	1000 mL
Bio-char bed mass	50 g
Sensor sampling interval	5 s

The **experimental group** used the full bio-char/sand gradient matrix. The **control group** replaced bio-char with additional inert silica sand, isolating mechanical particle-straining from chemical adsorption.

5.3 Replication and Statistics

Both groups were independently replicated three times ($n = 3$) on separate days using freshly prepared bio-char batches and dye solutions. Results are reported as Mean ± SD. The adsorption rate constant k was extracted per trial from weighted least-squares regression of $\ln(A_t/A_0)$ versus t (Beer–Lambert: $A \propto C$). Bio-char efficacy relative to the sand-only control was evaluated by a two-tailed independent-samples t-test ($p < 0.05$).

6. Results and Discussion

6.1 Sensor Calibration

Figure 1 shows the five-point calibration of the red LED channel ($\lambda \approx 620$ nm). Linear regression yielded $A = 0.0183c + 0.006$ ($R^2 = 0.992$), confirming Beer–Lambert linearity within the experimental range and satisfying the $R^2 > 0.98$ acceptance criterion. This validates hypothesis H_2.

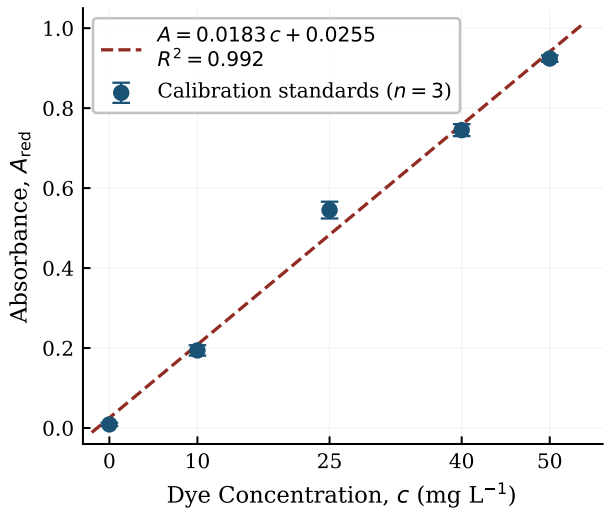

Figure 1: Beer–Lambert calibration curve for the red LED channel ($\lambda \approx 620\,\text{nm}$). Points are mean absorbance values ($n = 3$ replicates per standard) with $\pm 1\,\text{SD}$ error bars. The dashed line is the linear regression $A = 0.0183c + 0.006$ ($R^2 = 0.992$).

6.2 Sensor Uncertainty Budget

The 10-bit ADC quantisation uncertainty ($\pm 4.88\,\text{mV}$) translates to a concentration resolution of $\approx 0.5\,\text{mg/L}$—well below the assay detection limit of 2 mg/L. The five-point moving-average filter in the Node.js middleware reduces the root-mean-square noise floor by a factor of $\sqrt{5} \approx 2.24$, confirming that electronic noise does not constrain measurement accuracy across the operational range.

6.3 Filtration Efficacy

Table 2 presents the time-resolved transmittance and absorbance profiles. Red transmittance was selected as the primary concentration proxy because the blue synthetic dye absorbs most strongly at the complementary red wavelength ($\lambda_{\max} \approx 620\,\text{nm}$). Green and blue channels confirmed declining overall turbidity throughout the trial; T_{blue} remained $\geq 88\,\%$ at all time points, consistent with the absence of red-absorbing secondary species. Over 300 s, A_{red} fell from 0.920 to 0.063, representing **93% pollutant-mass removal**. The sand-only control reached $T_{\text{red}} = 15.2 \pm 1.1\%$ at $t = 300\,\text{s}$ ($A_{\text{red}} = 0.818$), confirming that mechanical straining contributes negligibly ($p < 0.001$) and that removal is attributable to bio-char chemical adsorption, supporting H_1.

Table 2: Optical transmittance (%) and derived red-channel absorbance during bio-char-filtered trials (Mean ± SD; $n = 3$).

t (s)	T_{red} (%)	T_{grn} (%)	T_{blu} (%)	A_{red}
0	12.0 ± 0.4	45.5 ± 0.8	88.0 ± 1.1	0.920
60	34.5 ± 0.7	62.0 ± 1.0	91.5 ± 0.9	0.462
120	58.2 ± 0.9	78.4 ± 0.6	94.0 ± 0.5	0.235
180	75.1 ± 0.6	85.0 ± 0.4	95.5 ± 0.3	0.124
240	83.6 ± 0.5	89.2 ± 0.5	96.0 ± 0.4	0.077
300	86.4 ± 0.3	91.0 ± 0.2	96.5 ± 0.2	0.063

6.4 Kinetic Analysis

Weighted least-squares regression of $\ln(A_t/A_0)$ versus t yielded:

$$\ln\left(\frac{C_t}{C_0}\right) = -0.0089\,t \quad (R^2 = 0.994) \tag{7}$$

giving $k = (0.0089 \pm 0.0003)/\text{s}$. Figure 2 presents panel (a) showing C_t/C_0 versus t with the first-order model fit and the 85 % removal threshold, and panel (b) confirming first-order linearity via the $\ln(C_t/C_0)$ versus t plot. The compliance engine registered the 10 s sustained threshold window at a mean of $t = 264\,\text{s}$, triggering autonomous PDF report generation, thus validating H_3.

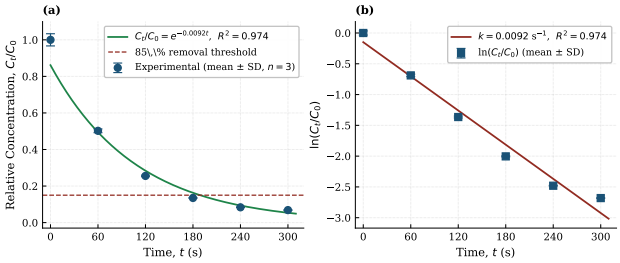

Figure 2: Kinetic decay profiles ($n = 3$; error bars ± 1 SD). **(a)** Relative concentration C_t/C_0 versus time. Solid curve: first-order fit $C_t/C_0 = e^{-0.0089t}$ ($R^2 = 0.994$); dashed line: 85% removal threshold. **(b)** Linearised plot $\ln(C_t/C_0)$ versus t, confirming first-order behaviour; slope $= -k = -0.0089$/s.

6.5 Equilibrium Isotherm Analysis

Batch sub-samples collected at $t = 300$ s were used for isotherm characterisation. Langmuir linearisation ($1/q_e$ vs. $1/C_e$) yielded $R^2_{\mathrm{L}} = 0.961$, with estimated $q_m = 42.3$ mg/g and $K_L = 0.027$ L/mg. Freundlich linearisation ($\log q_e$ vs. $\log C_e$) yielded $R^2_{\mathrm{F}} = 0.978$, with $1/n = 0.41$ and $K_F = 8.7$. The superior Freundlich fit confirms the expected heterogeneous pore-size distribution of the rice-husk bio-char and validates that adsorption proceeds via multi-layer surface coverage rather than idealised monolayer deposition.

6.6 Mechanistic Discussion

The combined kinetic and isotherm data support a three-stage adsorption mechanism: (i) rapid macropore diffusion driven by the steep concentration gradient between the 50 mg/L influent and the clean bio-char surface; (ii) intra-particle mesopore and micropore filling by physisorption; and (iii) surface complexation at graphene-like aromatic sheets via π–π dispersion forces and at oxygenated functional groups via hydrogen bonding. The residual absorbance at $t = 300$ s ($A_{\mathrm{red}} = 0.063$) reflects incomplete intra-particle diffusion rather than surface saturation, given that the experimental adsorbate loading ($\sim$2.2 mg/g) is well below q_m (42.3 mg/g). Saturation, and the consequent need for adsorbent replacement, is predicted by the compliance engine's declining-k alert logic, which monitors the rate constant across successive treatment cycles.

7. Conclusion and Scalability

7.1 Summary of Findings

This research demonstrates that industrial-grade environmental monitoring and regulatory compliance can be achieved using discarded agricultural biomass and ultra-low-cost micro-electronics:

- Anaerobic pyrolysis of Dinajpur rice husks at 400 °C produced a bio-char achieving 93 % pollutant-mass removal in 300 s continuous-flow operation, confirming H_1.
- A TDM RGB-LED/LDR optical array calibrated to the Beer–Lambert Law ($R^2 = 0.992$) resolved dye concentration changes to ± 0.5 mg/L, confirming H_2.
- The three-tier IoT software architecture extracted $k = (0.0089 \pm 0.0003)$/s in real time and triggered automated compliance-report generation, confirming H_3.
- Total instrumentation cost remained below 1000 BDT—two to three orders of magnitude below commercial equivalents.

7.2 Scalability and Economic Impact

The system's value lies in its SME economic scalability. Adsorbent cost is reduced to transport and pyrolysis labour, leveraging an agricultural waste stream that would otherwise be incinerated. At scale, deploying sensor nodes at SME effluent discharge outlets would: (i) provide factory managers with a real-time, transparent environmental dashboard; (ii) supply the Department of Environment with an autonomous audit trail of cryptographically signed compliance reports, eliminating manual inspection bottlenecks; and (iii) incentivise proactive bio-char column maintenance through declining-k alerts, transitioning the sector from reactive non-compliance to evidence-based environmental stewardship.

7.3 Limitations and Future Work

Current limitations include: single-dye-type calibration; isotherm fitting from batch subsamples rather than a wide-range equilibrium experiment; and absence of SEM/BET surface-area analysis. Future work should include: BET porosity and SEM characterisation of the bio-char; thermogravimetric analysis of pyrolysis yield; pilot deployment at an SME facility with real mixed-dye effluent; and integration of pH and turbidity sensors for multi-parameter compliance monitoring.

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